

Working draft: the 540-run panel is complete; the nominal-contract study and the abstract are pending. This reading copy is not an ICRA template or a submission-compliance check.

Motion2Scene: Constructing Training Scenes from Executed Humanoid Motion Contrasts

Working manuscript, September 7, 2026. Abstract intentionally deferred until the registered result decision. This draft separates completed measurements from the unfinished M2S-ICRA-v1 comparison. It is not a submission-ready paper.

I. Introduction

A humanoid that can walk and crouch still needs to decide when crouching helps. Training this decision requires encounters whose observations distinguish the outcomes of the available commands. A randomly placed obstacle can be irrelevant to both commands, or defeat both. A beam placed above a crouching reference can appear useful while colliding with the robot during its actual transition into the crouch. The resulting label concerns an intended posture rather than the command the robot can execute from its current state.

We study scene construction from a pair of supported humanoid commands: commit to walking, or request a crouching reference at a fixed decision time. Both commands run through the same frozen controller. Their achieved trajectories provide a proposal model for placing overhead constraints, and paired obstacle-present simulations provide outcome labels. A fixed perceptive selector then predicts the success of each command. This separates three questions: whether a scene has a geometric contrast, whether that contrast survives execution, and whether the resulting data teaches a useful decision on other layouts.

Learning environments from motion is not itself our novelty claim. Learning from Learned Hallucination (LfLH) already learns obstacle configurations from open-space motion and uses hallucinated environments to train navigation policies [1]. Our controlled instance addresses a different supervision problem: full-body humanoid commands must enter and leave an adaptation through a legal transition, and an obstacle may change contact outcomes without making progress physically impossible.

The study makes three contributions at distinct evidence levels. First, it defines a command-bound construction and labeling protocol that preserves all four paired outcomes, including encounters that neither command solves. Second, it measures the mismatch between complete-reference

screening and achieved-command screening, and validates source-specific switching banks and physical contrasts in Isaac Lab. Third, it specifies a four-arm comparison—uniform, analytic, target-only and learned construction—with one shared outcome learner. The third contribution remains an experimental question, and the completed panel answers part of it: contrast construction helps, and the learned proposal supplied nothing to test. The learned proposal is a tested factor, not an assumed source of superiority.

The experiment's most transferable outcome is about scarcity rather than about which generator wins. A training scene helps this decision only if it carries a contrast that survives execution and is visible when the command must be issued, and such scenes are rare in a way that is measurable before any simulation: the acceptance geometry alone can remove almost all of them, and we report the price of that choice as a curve rather than a single threshold. On the other side, scarcity cuts both ways. A single verified contrast is enough to change what the fixed learner does, which is what the removal diagnostic in Section VI shows.

II. Related work

LfLH learns hallucinated environments in which open-space plans are useful and trains reactive navigation from those environments [1]. This establishes the motion-to-scene-to-policy loop that motivates our experiment. We investigate how paired executed humanoid commands constrain the labels needed by that loop, rather than claiming that inverse scene generation is new.

Recent humanoid systems address perception-conditioned traversal and skill composition at broader scales. HumanoidPF represents humanoid–obstacle relationships for cluttered traversal [2]. Perceptive Humanoid Parkour composes atomic skills with motion matching and studies long-horizon vision-based execution [3]. Our experiment retains a narrow beam family and command bank to isolate training-scene construction. It does not compare against those systems' full task breadth or sensing pipelines.

SONIC supplies the inherited motion-tracking foundation and command interface [4]. We freeze its parameters and do not attribute its locomotion capacity to scene construction. The learning component is a supervised two-head outcome

predictor. Because its command is issued once from a shared approach, the present task is not an end-to-end reinforcement-learning study. Sequential extensions would need to address policy-induced state distributions, as emphasized by DAGger [5]. Evaluation reports paired outcomes and the small number of carrier groups, consistent with the need to expose uncertainty rather than infer robustness from point estimates [6].

III. Problem and command-bound supervision

Let S denote a static scene, x_t the measured robot state, ϕ_t the reference phase, h_t the relevant approach history, and a a supported command. Define the binary task outcome as $Y_a = Y(S, x_t, \phi_t, h_t, a)$. A0 commits to walking through the encounter. A1 requests d040 at $t=0.30$ s and uses the shared legal return interface. A0 is not a wait-and-reconsider action. A refusal from the predictor continues A0; it is not a physical stop or successful avoidance.

The contact-qualified task requires at most 1 N measured beam force through passage, body-origin crossing 0.1 m beyond the beam, and 0.3 s upright stability in the first episode. We record resets, falls, legal entry/return and endpoint tracking separately. This definition admits contact-qualified passage with an endpoint-tracking rejection; it does not imply exact reference completion or an all-time collision guarantee.

Each encounter receives the pair (Y_0, Y_1) . Both-pass examples permit a preference for walking. Walk-fail/crouch-pass examples establish useful adaptation. Walk-pass/crouch-fail examples warn against the transition. Both-fail examples describe the limits of the current command bank. Missing or mismatched executions are masked rather than converted to negative labels. Paired executions must agree on recorded pre-decision state, sensing, action and token histories. This is measured prefix agreement, not a claim to restore an unobserved complete controller state.

IV. Construction and fixed learner

For each qualified carrier, empty-scene executions produce achieved walk and d040 body trajectories. A capsule representation supplies inexpensive geometric queries. The station coordinate follows the source reference route; beam dimensions are 0.10 by 1.20 by 0.10 m, with station in $[0.1, 0.9]$ and underside in $[1.1, 1.45]$ m. The achieved forecast includes entry, passage and exit. Physics remains the labeling oracle because obstacle interaction and tracking deviations can invalidate that forecast.

Uniform construction samples the common domain without a future-trajectory contrast filter. Analytic construction uses achieved envelopes and global distinct search. Motion2Scene

retains the frozen local-event initializer and 17-query bounded pattern search, replacing its geometric evaluator with achieved trajectories. The target-only ablation uses its previously fitted no-contrast model and removes walking interference from search and acceptance. All arms share finite-domain, duplicate, reserved-layout and pre-decision validity checks. Target and contrast screens retain the declared perturbation offsets; every rejected attempt remains charged.

The observation contains 144 values from twelve upper rays and their conditional lower-ray queries, plus 70 state/phase/history values. Neither source identity nor true beam coordinates enters a learned selector. One deterministic logistic model predicts two probabilities from the 214 inputs. All inputs have physical scale one except joint velocity, scaled by five. Zero initialization, 2000 Adam updates at 0.01 and a 0.01 mean-squared-weight penalty are common across arms. The ≥ 0.5 rule prefers walking when both heads are positive and requests d040 only when walking is predicted infeasible and d040 feasible. The transition manager enforces legality; it does not make a hidden scene-dependent choice beneath the model.

V. Completed physical and mechanism evidence

The earlier generated-source pilot admits six source pairs from eight requested and obtains eighteen contact-avoidance contrasts from twenty-four requested scene slots. Walking still crosses in those cases but violates the contact criterion. Thus the result distinguishes task outcomes, not physical impossibility. d040 also passes the selected 41002 beam in three runs, so d055 is not established as minimally necessary. [Internal evidence: SOURCE_EXECUTION_V1_RESULT, BEAM_D040_EXECUTION_V1_RESULT.]

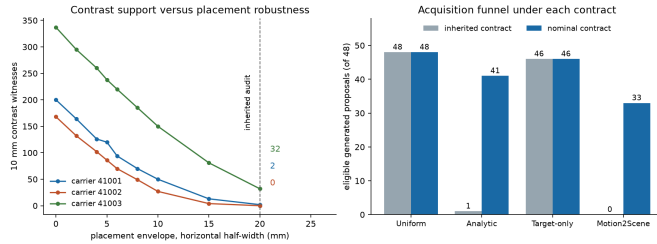
The selector-breakpoint study identifies one d040 rescue among six matched layout/seed conditions on carrier 41002. A shared analytic-trained linear control realizes this rescue: at underside 1.27 m and seed 8512, d040 records 0 N through qualifying passage, where matched walking records 57.947 N. Its 4/6 passage contrasts with 3/6 for the other three data arms. This is development evidence on one inspected carrier. Changing scaling, capacity and regularization together does not isolate normalization as the sole causal explanation of the original MLPs' failure.

The achieved-transition forecast catches thirteen command contacts that the complete- reference screen falsely predicts clear, but misses two other contact-qualified failures. It improves a measured diagnostic and remains imperfect. The next empty- command bank qualifies 41001, 41002 and 41003 in both seeds: twelve executions, six paired approaches and legal d040 entry/return. Qualification does not require every reference-tracking metric to pass.

The first bounded construction pilot accepts only one of twelve assigned slots, from analytic construction on 41001; no learned slot survives. Both seeds then show walking contact (1635.077 and 1130.936 N) and d040 contact-qualified passage (0 N). Both d040 runs also receive reference_endpoint_tracking_error rejections. The full funnel therefore demonstrates one unique analytic scene, not broad learned-scene yield.

Why almost nothing is acceptable: the placement envelope consumes the window

The acceptance geometry, not the proposal model, is the dominant cause. Sweeping 11,421 station/height centres per carrier over the recorded achieved transitions and scaling the inherited audit envelope gives the number of 10 mm two-sided contrast witnesses below. The nominal-pose column is the achievable support; the last row is the envelope the study inherited.



Envelope (xy / z)	41001	41002	41003
nominal pose	200	168	337
+5 mm / +-2.5 mm	120	86	238
+10 mm / +-5 mm	50	27	150
+20 mm / +-10 mm (inherited)	2	0	32

The best nominal joint margin, the smaller of target clearance and walk interference at a centre's own pose, is 24.42, 20.48 and 26.99 mm. The inherited envelope asks a centre to keep a 10 mm two-sided contrast while the beam moves +-20 mm horizontally, which is the size of the entire executed window. It is the correct instrument for a placement certificate and for any hardware claim, where the obstacle's realized pose is uncertain, and the wrong one for a simulator study that authors the beam at an exact pose. Reported this way the table also prices a future hardware claim directly.

A second and separable cause belongs to the proposals. On 41003 the inherited envelope still leaves 32 sampled witnesses and the frozen learned draws, whose +-0.05 station and +-0.03 m trust boxes are unchanged, reach none of them. Reachability and acceptance therefore fail independently, and only the first is addressed by a different contract.

A third property is measured and not gated by either contract. Among proposals that are critical at the nominal pose, 44 of 47 analytic scenes but only 24 of 34 learned scenes are visible to the decision-time sensor, with 8 of 12 invisible on 41003, where the learned initializer concentrates late-station draws. A contrast the robot cannot observe cannot inform the selector whatever its geometry. No visibility gate is added after the fact; the quantity is reported as a covariate.

These are sampled witnesses on one finite grid at one physics seed per carrier. They do not prove infeasibility, do not estimate feasible volume, and do not revise any earlier outcome.

VI. Registered learning-utility experiment

M2S-ICRA-v1 assigns 24 encounter groups per arm across the three qualified development carriers: eighteen generated and six identical shared backgrounds. All four methods retain their refused assignments. At most 156 unique command executions label the four arms and shared backgrounds. If acquisition produces fewer complete groups, we report actual counts and treat the result as an equal-requested-slot comparison; we cannot claim to have completed an equal-24-label experiment.

Four fixed learners and two declared comparators face twelve reserved station/height layouts, two physics seeds and three carriers, yielding 432 traversal assignments. The comparators are the upper/lower-ray decision rule and a privileged achieved- geometry predictor, not an optimal transition-time oracle. Passage is averaged within carrier before averaging carriers. Paired tables report both-pass, method-A-only, method-B-only and both-fail. Repeats share carriers and do not create independent source evidence. Background adaptation/refusal outcomes form separate suites.

All 82 assigned labeling executions and 41 unique pairs pass measurement admission. The resulting training sets are the study's central observation.

Arm	Groups	Generated	Both pass	d040 only	Walk only	Both fail
Uniform	24	18	10	0	0	14
Analytic	7	1	4	1	0	2
Target-only	22	16	20	0	0	2
Motion2 Scene	6	0	4	0	0	2

Every arm contains the same six shared background groups. Across 192 proposals, 72 assigned generated slots and 82 executed label commands, **exactly one** acquired group is a walk-fail/d040-pass contrast, from analytic construction on 41001. Consequently the analytic and Motion2Scene training

sets differ by exactly that one group and are otherwise identical, and the equal-24-complete-label target fails in three arms. This is an equal-requested-slot comparison with unequal acquired data, and the acquisition shortfall must be disclosed wherever the downstream numbers are quoted.

The comparison this creates is unusually clean and unusually narrow at the same time. Adding eighteen untargeted groups (uniform) or sixteen target-only groups changes the fitted policy not at all; adding one contrast changes it completely. Both statements rest on a single acquired example, so the effect size is not estimable from this study.

Four primary fits and 24 registered leave-four-assignment-out refits are complete. All 540 policy evaluations are admitted: six methods on all 72 traversal conditions, plus the 108 background control runs. Passage is uniform 28/72, analytic 39/72, target-only 28/72, background-only Motion2Scene 28/72, scripted rays 46/72 and privileged geometry 39/72. Nothing remains pending.

Analytic versus uniform yields 28 both-pass, eleven analytic-only, zero uniform-only and 33 both-fail conditions. The carrier-averaged difference is +15.278 percentage points, with six additional passes on 41001, three on 41003 and two on 41002. All eleven additional passages request d040 and measure 0 N beam force; matched walking records 385.1–1685.5 N. This supports useful learned adaptation on the inspected layouts. It does not establish a learned-generator advantage or source-held-out generalization.

The comparison is one contrast against three copies of the same policy, and must be reported as such. The uniform, target-only and background-only Motion2Scene learners issue walking on all 72 conditions and are behaviourally identical; each therefore gives the same 28/72 and the same eleven-to-zero paired table against analytic. The condition-level discordance test over eleven one-directional pairs gives $p = 0.00098$, but the 72 conditions are twelve layouts and two seeds inside only three carriers, so that figure overstates independence. The carrier-level statement is the honest one: the direction is the same on all three carriers, with six, three and two extra passes and no reverse case, which is $n = 3$.

What the single contrast taught is legible, and it is narrow. That group's beam sits at underside 1.2755 m. The evaluation panel has three underside bands, and the fitted learner's behaviour separates along them.

Beam underside	Conditions	Analytic requests d040	Analytic passes	Scripted requests	Walking-only passes
1.18 m	24	11	0	11	0
1.27 m	24	14	15	23	4
1.36 m	24	4	24	12	24

The learner almost never requests the adaptation at 1.36 m, where walking already passes 24 of 24, and all eleven of its additional passages fall in the 1.27 m band, the band its one training contrast came from, spread across all three carriers with six on 41001. At 1.18 m neither command ever succeeds, so its eleven requests there are neither rewarded nor punished by this suite. Within the 1.27 m band it requests the adaptation 14 times in 24 conditions against the scripted rule's 23, which is where its remaining gap to the observed ceiling lies.

So one example transferred across carriers within its own geometry band and not beyond it. That is a real but strictly band-local generalization, and it is the most this study can claim about what an acquired contrast teaches.

Adaptation is close to free on this panel and every learner under-uses it. Of the 47 conditions in which both commands were actually executed, 19 are d040-only successes, 16 both-pass, 12 both-fail, and none is a walking-only success. The best outcome available from the observed commands is 47 of 72, which the scripted rule nearly attains at 46 by requesting d040 46 times; analytic requests it 29 times and reaches 39. Under-adaptation, not misfiring adaptation, is what separates the learners from the ceiling here.

The background suites settle the cost side. Across the absent and raised controls no policy requests the adaptation even once, so no arm pays an unnecessary-adaptation penalty on this panel and the learned request is not indiscriminate. The blocked suite reverses the ranking: every learner and the privileged forecast classify the scene as infeasible and refuse on 6 of 6, while the scripted rule refuses on none. No blocked scene is passable, so refusal there is correct classification rather than successful avoidance, and it is reported apart from passage. The stronger traversal baseline is therefore the weaker infeasibility detector, and neither dominates.

The first analytic refit removes the sole useful generated label and three refused assignments. Its remaining training IDs, weights, biases and scales exactly equal the background-only Motion2Scene primary model. On the 61 recorded inputs, its d040 requests fall from 22 to zero. The five other analytic folds retain 22 requests; two remove no available labels. This controlled removal links one acquired contrast to the fixed learner's decisions. It is an offline refit diagnostic, not additional physics evaluation or a population claim about single-example learning.

Acquisition spends 0.824108 contended GPU h and admitted policy evaluations spend 3.146765 h. These are physical execution costs, not a complete end-to-end cost comparison: generator fitting, proposals, rejected searches and shared bank costs remain separately recorded. Remaining evaluation and the final grouped uncertainty analysis must precede the registered result decision. [Internal evidence:

M2S_ICRA_540_RESULT; all assigned outcomes and refit checkpoints are released.]

A second contract, registered before its proposals

Because the envelope and not the proposal model dominates the funnel, a second contract is registered separately, with its predictions filed before any proposal was drawn. It changes exactly one thing: criticality is judged at the nominal pose, with the 17-offset and 113-offset survivals still computed and stored for every proposal as reported diagnostics that never gate acceptance. Everything else is unchanged, and the six shared background groups are reused by reference rather than re-executed, so no reused trace counts as a new example. A unit test asserts that both contracts reach identical decisions whenever their geometry verdicts agree.

Under this contract eligible generated proposals move from 1 of 48 to 41 of 48 for analytic and from 0 of 48 to 33 of 48 for Motion2Scene, with at least one eligible group on every carrier for both. Three groups per carrier and arm are assigned, giving 36 groups and 72 label commands. Of the 18 assigned contrast-arm proposals, 17 fail the 113-offset diagnostic, which is the expected consequence of the measured envelope price rather than a new finding. Larger accepted yield is geometric acceptance only: whether these are useful contrasts is decided by the paired physical labels, and whether they teach is decided by the evaluation. Both are pending at this snapshot, and no learning conclusion is drawn from acceptance counts.

VII. Limitations and result decision

This is a controlled instance with one beam family, two commands, ideal simulator rays, three inspected development carriers and one frozen controller. It has no hardware or source-held-out transfer claim. The original 120/600 MLP evaluation remains a retained partial failure, with 480 assignments paused. Existing layout diagnostics have

informed this revision, so their reuse is development evaluation.

Geometric proposal acceptance does not guarantee a useful physical contrast. The current learned initializer has produced no accepted scenes in the achieved-transition study, and the matched 24-label quota fails in three arms. Contact-qualified passage does not eliminate endpoint tracking error. Neither a predicted refusal nor continued neutral walking is a qualified protective action. The final conclusion must follow the registered comparison: report contrast-construction utility if supported, learned proposal benefit only if measured, and a construction/validation result if learning utility remains unresolved.

References (verified primary records, September 7)

- [1] Z. Wang et al., “From Agile Ground to Aerial Navigation: Learning from Learned Hallucination,” IROS, 2021.
<https://arxiv.org/abs/2108.09793>
- [2] H. Xue et al., “Collision-Free Humanoid Traversal in Cluttered Indoor Scenes,” arXiv:2601.16035, 2026.
<https://arxiv.org/abs/2601.16035>
- [3] Z. Wu et al., “Perceptive Humanoid Parkour: Chaining Dynamic Human Skills via Motion Matching,” arXiv:2602.15827, 2026. <https://arxiv.org/abs/2602.15827>
- [4] Z. Luo et al., “SONIC: Supersizing Motion Tracking for Natural Humanoid Whole-Body Control,” Science Robotics, vol. 11, no. 117, eaed4592, 2026.
<https://arxiv.org/abs/2511.07820>
- [5] S. Ross, G. Gordon, and D. Bagnell, “A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning,” AISTATS, 2011.
<https://proceedings.mlr.press/v15/ross11a.html>
- [6] R. Agarwal et al., “Deep Reinforcement Learning at the Edge of the Statistical Precipice,” 2021.
<https://arxiv.org/abs/2108.13264>